

NEIL'S ALGEBRA 1 CHEAT SHEET (page 1 of 2)

Quadratics core. Print double-sided with page 2.

LINEAR EQUATIONS

Slope between two points

$$m = (y_2 - y_1) / (x_2 - x_1)$$

Slope-intercept form

$$y = mx + b$$

m = slope, b = y-intercept

Point-slope form

$$y - y_1 = m(x - x_1)$$

Standard form

$$Ax + By = C$$

Parallel & perpendicular

Parallel: same m. Perp: $m_1 \cdot m_2 = -1$

EXPONENT RULES

Product of powers

$$x^a \cdot x^b = x^{(a+b)}$$

Quotient of powers

$$x^a / x^b = x^{(a-b)}$$

Power of a power

$$(x^a)^b = x^{(a \cdot b)}$$

Zero & negative

$$x^0 = 1 \quad x^{-a} = 1 / x^a$$

Power of a product

$$(xy)^a = x^a \cdot y^a$$

Rule: same base -> combine exponents

FACTORING PATTERNS

Step 1 ALWAYS: GCF first

$$6x^2 + 9x \rightarrow 3x(2x + 3)$$

Difference of squares

$$a^2 - b^2 = (a + b)(a - b)$$

Perfect square trinomial

$$a^2 + 2ab + b^2 = (a + b)^2$$

Trinomial $x^2 + bx + c$

Find two numbers: product = c, sum = b

Trinomial $ax^2 + bx + c$ ($a > 1$)

AC-method: product $a \cdot c$, sum b, split bx

QUADRATIC FORMS & GRAPHING

Standard form

$$y = ax^2 + bx + c$$

Vertex form

$$y = a(x - h)^2 + k$$

Vertex is at (h, k). Sign flip on h!

Factored form (x-intercepts)

$$y = a(x - r_1)(x - r_2)$$

Axis of symmetry

$$x = -b / (2a)$$

Direction / width

$a > 0$ opens up, $a < 0$ opens down

$|a| > 1$ narrow, $|a| < 1$ wide

SOLVING QUADRATICS -- 4 METHODS

1. Factoring

Set = 0, factor, zero-product rule

2. Square roots

$$\text{if } x^2 = k, \quad x = \pm \sqrt{k}$$

3. Completing the square

Move c. Add $(b/2)^2$ to BOTH sides

4. Quadratic formula (ALWAYS works)

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Pick the method

Factorable? Factor. $x^2 = \#?$ Sqrt.

DISCRIMINANT + WORD PROBLEMS

Discriminant

$$D = b^2 - 4ac$$

$D > 0$: two real solutions

$D = 0$: one real solution (double)

$D < 0$: no real solutions

Projectile motion (ft, sec)

$$h(t) = -16t^2 + v_0 \cdot t + h_0$$

Max height at

$$t = -b / (2a) \quad (\text{axis of sym})$$

Hits ground when

$$h(t) = 0 \quad (\text{solve for } t, \text{ take } t > 0)$$

When stuck: 1) Write what's given. 2) Write what you want. 3) Draw it. 4) Pick ONE method and start. If nothing works, try the quadratic formula -- it

NEIL'S ALGEBRA 1 CHEAT SHEET (page 2 of 2)

Print double-sided. All the rest of Algebra 1.

INEQUALITIES

Solve like equations, BUT...

Flip the sign if you mult/div by negative

$$-2x < 6 \rightarrow x > -3 \quad (\text{flip!})$$

Compound AND (between)

$$-2 < x + 1 < 5$$

Solve both parts together

Compound OR

$$x < -3 \quad \text{OR} \quad x > 5$$

Absolute value

$$|x| < a \rightarrow -a < x < a$$

$$|x| > a \rightarrow x < -a \quad \text{OR} \quad x > a$$

SYSTEMS OF EQUATIONS

Substitution

Solve one eq for a var, sub into other

$$y = 2x + 1 \quad 3x + y = 16$$

$$\rightarrow 3x + (2x+1) = 16 \rightarrow x = 3, y = 7$$

Elimination

Add/subtract to cancel a variable

Line up like terms; scale to match

One/None/Infinite solutions

Intersect once $\rightarrow$ one. Parallel $\rightarrow$ none.

Same line $\rightarrow$ infinitely many.

FUNCTIONS & NOTATION

Is it a function?

Each x has exactly one y

Vertical line test on a graph

Function notation

$$f(x) = 2x + 3$$

$$f(4) = 2(4) + 3 = 11$$

Domain & range

Domain = all x -values (inputs)

Range = all y -values (outputs)

Evaluating with variables

$$f(x+1) = 2(x+1) + 3 = 2x + 5$$

RADICALS & SQUARE ROOTS

Simplify by factoring

$$\sqrt{48} = \sqrt{16 \cdot 3} = 4\sqrt{3}$$

Multiply/divide

$$\sqrt{a} \cdot \sqrt{b} = \sqrt{a \cdot b}$$

$$\sqrt{a} / \sqrt{b} = \sqrt{a/b}$$

Add/subtract (like terms only)

$$3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$$

Rationalize denominator

$$1/\sqrt{3} = \sqrt{3}/3$$

Multiply top & bottom by $\sqrt{3}$

PYTHAGOREAN + DISTANCE

Pythagorean theorem (right tri.)

$$a^2 + b^2 = c^2$$

c is the hypotenuse (longest side)

Example

$$a=3, b=4 \rightarrow c = \sqrt{25} = 5$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$$

Slope (for reference)

$$m = (y_2 - y_1)/(x_2 - x_1)$$

VARIATION + WORD PROBLEMS

Direct variation

$$y = kx \quad (y/x = \text{constant})$$

If $y=12$ when $x=4 \rightarrow k=3 \rightarrow y=3x$

Inverse variation

$$y = k/x \quad (x \cdot y = \text{constant})$$

If $y=3$ when $x=8 \rightarrow k=24 \rightarrow y=24/x$

Word-problem steps

1) Define variable clearly

2) Translate words $\rightarrow$ equation

3) Solve. 4) Check: does it fit story?

Key habits: Show your work. Check by plugging back in. Units matter in word problems. Watch negative signs with inequalities!